DAY 24 – SECURITY EVENT CLASSIFICATION

Cyber Security Internship – Veda Technology

1. Objective

To understand the fundamentals of security event classification and develop basic security triage skills by analyzing sample security logs. The task focuses on categorizing events as normal, suspicious, or potentially malicious based on their characteristics, frequency, and possible security impact.

2. Introduction

Security event classification is the process of analyzing system and network activities to determine whether they represent normal operations or possible cybersecurity threats.

Organizations generate numerous security events through authentication systems, firewalls, applications, and network devices. Not every event indicates a security incident.

Security analysts examine these events to identify unusual behavior, assess potential risks, and determine whether further investigation is required.

Security triage helps prioritize events according to their severity, potential impact, and available evidence.

Categories of Security Events

Classification	Description	Example
Normal	Expected and authorized activity	Successful employee login
Suspicious	Unusual activity requiring investigation	Multiple failed login attempts
Potentially Malicious	Activity strongly suggesting an attack or unauthorized action	Detected malware execution

3. Tools and Technologies

Tool	Purpose
Microsoft Excel	Organizing and classifying security events
Google Sheets	Reviewing and documenting security logs
Filters	Separating events by classification
Conditional Formatting	Highlighting potentially risky events
Pivot Tables	Summarizing event classifications
Synthetic Dataset	Simulating realistic security activities

4. Dataset Description

A synthetic dataset containing 15 security events was prepared for classification.

The dataset includes authentication activities, firewall events, file access attempts, malware alerts, and system operations.

Each event was reviewed based on its behavior and potential security implications.

Sample Security Event Dataset

Event ID	Security Event	Classification
E01	Successful employee login	Normal
E02	Scheduled system backup	Normal
E03	Authorized software update	Normal
E04	Successful administrator login	Normal
E05	Routine firewall traffic	Normal
E06	Multiple failed login attempts	Suspicious
E07	Login from an unusual location	Suspicious
E08	Unexpected access to restricted files	Suspicious
E09	Repeated connection attempts	Suspicious
E10	Login outside regular working hours	Suspicious
E11	Malware execution detected	Potentially Malicious
E12	Confirmed unauthorized file modification	Potentially Malicious
E13	Credential-stealing activity detected	Potentially Malicious
E14	Confirmed malicious command execution	Potentially Malicious
E15	Unauthorized data transfer detected	Potentially Malicious

5. Security Event Classification Criteria

Security events were categorized based on observed behavior, event frequency, and possible security impact.

Criteria	Normal	Suspicious	Potentially Malicious
User Behavior	Expected	Unusual	Unauthorized or harmful

Login Activity	Authorized login	Repeated failures	Confirmed credential attack
File Access	Authorized	Unexpected	Confirmed unauthorized modification
Network Traffic	Expected traffic	Unusual connections	Confirmed malicious communication
Security Impact	Low	Requires assessment	Significant potential impact
Response	Routine monitoring	Investigation	Urgent investigation and possible containment

6. Practical Implementation

Step 1: Create a Security Event Dataset

Microsoft Excel or Google Sheets is used to create a spreadsheet containing synthetic security events.

The dataset includes the following columns:

Column	Description
Event ID	Unique event identifier
Timestamp	Date and time of occurrence
Username	Account associated with the event
Source IP	Network address associated with the event
Event Description	Details of the activity
Classification	Normal, Suspicious, or Potentially Malicious
Reason	Explanation supporting the classification

Step 2: Analyze Individual Security Events

Each event is examined to identify whether the activity is expected, unusual, or indicative of malicious behavior.

For example, a successful login during normal working hours may be classified as normal when it matches authorized user behavior.

Repeated failed login attempts from the same source may be classified as suspicious because they could indicate password-guessing activity.

A confirmed malware execution event is classified as potentially malicious because it indicates harmful activity requiring immediate investigation.

Step 3: Assign Event Classifications

The events are categorized into three groups:

- Normal: Expected activities with no identified security concerns.
- Suspicious: Unusual events that require additional verification.
- Potentially Malicious: Events with strong evidence of harmful or unauthorized activity.

Step 4: Document Classification Reasons

The reason for each classification is recorded to support security analysis and provide a clear explanation of the decision.

Event ID	Classification	Reason
E01	Normal	Successful login consistent with authorized activity
E02	Normal	Backup performed according to schedule
E03	Normal	Approved system maintenance activity
E04	Normal	Administrator login matches authorized access
E05	Normal	Traffic matches established firewall rules
E06	Suspicious	Repeated failures may indicate password guessing
E07	Suspicious	Login originates from an unexpected location
E08	Suspicious	Restricted file access is inconsistent with expected behavior
E09	Suspicious	Repeated connections may indicate scanning
E10	Suspicious	Login timing differs from usual activity
E11	Potentially Malicious	Security alert indicates malware execution
E12	Potentially Malicious	Unauthorized modification has been identified
E13	Potentially Malicious	Activity indicates attempted credential theft
E14	Potentially Malicious	Malicious commands were detected

E15	Potentially Malicious	Unauthorized data transfer indicates possible data theft

Step 5: Apply Conditional Formatting

Excel or Google Sheets conditional formatting can be used to highlight classifications.

Classification	Suggested Highlight
Normal	Green
Suspicious	Yellow
Potentially Malicious	Red

This makes it easier to identify high-priority events during security monitoring.

Step 6: Summarize Classification Results

A summary table is prepared to calculate the number of events in each category.

Classification	Number of Events	Percentage
Normal	5	33.33%
Suspicious	5	33.33%
Potentially Malicious	5	33.33%
Total	15	100%

The dataset contains an equal number of events in each category to demonstrate all three classification levels.

7. Security Triage and Priority Assessment

Security triage involves evaluating events and deciding which activities require further investigation.

Event Category	Suggested Priority	Recommended Response
Normal	Low	Continue routine monitoring
Suspicious	Medium	Investigate event context and related activity
Potentially Malicious	High	Investigate urgently and consider containment

Priority should also consider affected assets, confidence in the evidence, and potential business impact.

Example Triage Workflow

1. Collect security event information.
2. Examine event details and supporting evidence.

3. Identify unusual or unauthorized activities.
4. Classify the event based on its characteristics.
5. Assign an appropriate investigation priority.
6. Document findings and recommend further action.

8. Security Risks and Recommendations

Identified Activity	Possible Risk	Recommended Action
Repeated Failed Logins	Password guessing	Configure login alerts and rate limiting
Unusual Login Location	Unauthorized account access	Verify identity and review login history
Restricted File Access	Data exposure	Review permissions and access records
Repeated Connections	Network scanning	Investigate traffic patterns
Malware Execution	System compromise	Isolate affected systems when warranted
Unauthorized Modification	Data integrity loss	Investigate changes and restore trusted data
Credential Theft	Account compromise	Revoke affected sessions and reset credentials
Unauthorized Data Transfer	Data leakage	Investigate destination and transfer details

9. Importance of Security Event Classification

Security event classification provides several benefits:

1. Threat Identification: Helps distinguish normal activities from potential security threats.
2. Efficient Investigation: Allows analysts to focus on events requiring attention.
3. Incident Prioritization: Helps determine which events should be investigated first.
4. Reduced Alert Fatigue: Prevents routine events from receiving unnecessary attention.
5. Security Monitoring: Supports continuous review of system and network activities.
6. Incident Response: Provides structured information for investigating and responding to threats.

10. Results and Observations

The synthetic security event classification exercise demonstrated how security activities can be organized into categories based on their characteristics.

Analysis Parameter	Result
Total Events Analyzed	15
Normal Events	5
Suspicious Events	5
Potentially Malicious Events	5
Classification Categories	3
Analysis Tools	Excel / Google Sheets
Main Finding	Unusual activities require contextual investigation
Recommended Approach	Risk-based security triage

Key Observations

- Normal events generally represent expected system activities.
- Suspicious events require additional analysis before determining whether an attack occurred.
- Potentially malicious events should receive greater investigation priority.
- Documenting classification reasons improves consistency and accountability.
- Security event classification helps analysts manage large volumes of security information.

11. Conclusion

The Security Event Classification task demonstrated the importance of analyzing security logs and categorizing events according to their potential security impact.

Using a synthetic dataset, 15 security events were classified as normal, suspicious, or potentially malicious. The exercise highlighted the importance of examining event behavior, documenting classification reasons, and assigning appropriate investigation priorities.

Excel and Google Sheets provide simple methods for organizing event records, applying filters, and summarizing classification results.

Overall, this task strengthened foundational security triage skills and demonstrated how systematic event classification supports threat detection, incident investigation, and effective cybersecurity monitoring.